

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

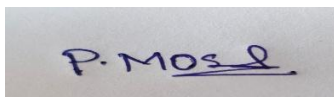
Analytical Problem Solving

Code of Conduct:

I P.MOSHIYA(192572142) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

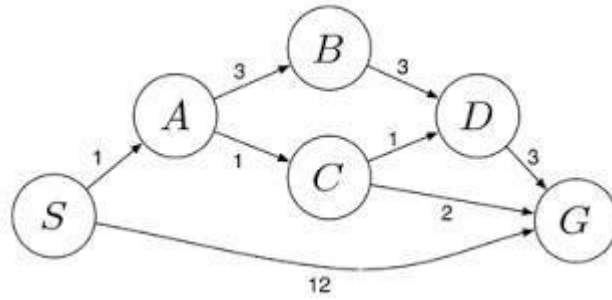
A photograph of a piece of paper with the handwritten signature "P. MOSHIYA" in blue ink.

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Criteria	Weightage	Q1 (9.4)	Q2 (9.4)	Q3 (9.4)	Q4 (9.4)	Q5 (9.4)	Total Marks (47)
Problem Understanding	20%	/1.88	/1.88	/1.88	/1.88	/1.88	/9.4
Method Selection	20%	/1.88	/1.88	/1.88	/1.88	/1.88	/9.4
Solution Process	25%	/2.35	/2.35	/2.35	/2.35	/2.35	/11.75
Accuracy of Calculation	20%	/1.88	/1.88	/1.88	/1.88	/1.88	/9.4
Interpretation of Results	15%	/1.41	/1.41	/1.41	/1.41	/1.41	/7.05
Total per Question	100%	/9.4	/9.4	/9.4	/9.4	/9.4	/47

ANSWER:

1. Solution Steps

Step	4-Gallon Jug	3-Gallon Jug	Action
1	0	0	Initial state
2	0	3	Fill the 3-gallon jug
3	3	0	Pour all water into the 4-gallon jug
4	3	3	Fill the 3-gallon jug again

Step	4-Gallon Jug	3-Gallon Jug	Action
5	4	2	Pour water into the 4-gallon jug until it is full
6	0	2	Empty the 4-gallon jug
7	2	0	Transfer the remaining 2 gallons into the 4-gallon jug

Final Answer

The 4-gallon jug now contains **exactly 2 gallons**.

2.i. Percepts

The Mars Rover gathers information from:

- Navigation cameras
- Panoramic cameras
- Temperature sensors
- Rock and soil composition sensors
- Obstacle detection sensors
- Position and orientation sensors
- Battery level
- Weather measurements

ii. Operating Environment

The environment is:

- Partially observable
- Dynamic
- Sequential
- Continuous
- Uncertain
- Single-agent
- Real-world environment

iii. Actions

The rover can:

- Move forward or backward
- Rotate left or right
- Avoid obstacles
- Capture images
- Drill rocks
- Collect soil samples
- Analyze collected samples
- Send scientific data to Earth
- Recharge using solar panels (when available)

iv. Performance Evaluation

The rover's success is measured by:

- Number of samples collected
- Accuracy of scientific analysis
- Safe obstacle avoidance
- Distance covered
- Energy efficiency
- Successful communication with Earth

- Mission completion

v. Suitable Agent Architecture

A Utility-Based Agent with Learning capability is the most suitable.

Reason:

- Chooses actions that maximize mission success.
- Learns from previous terrain conditions.
- Handles uncertain environments.
- Optimizes energy usage and safety.

3. Problem Components

Initial State

An empty chessboard with no queens placed.

State Space

Every valid arrangement where queens are placed column by column.

Actions

- Place one queen in any safe row of the next column.

Transition Model

Move from one configuration to another by placing one additional queen.

Goal Test

All eight queens are placed and:

- No two queens share the same row.
- No two queens share the same column.
- No two queens share the same diagonal.

Path Cost

Each move has equal cost.

Total cost equals the number of queens placed.

Search Strategy

The problem can be solved using:

- Backtracking Search
- Depth-First Search (DFS)
- Constraint Satisfaction Problem (CSP)

Backtracking is the most efficient classical approach.

4. Type of Problem-Solving Agent

A Goal-Based Agent.

Reason:

A Utility-Based Agent chooses the best option by considering different factors such as **cost, comfort, travel time, and customer preference**. In OLA, the customer can select different cab types like **Mini, Micro, Sedan, Shared, or Prime**, and the system helps choose the option that provides the highest satisfaction or utility.

Pseudocode (Steps):

1. Enter the pickup location.

2. Enter the destination.
 3. Display all available cab types.
 4. Customer selects a preferred cab.
 5. Check whether the selected cab is available.
 6. If available, calculate the fare and confirm the booking.
 7. Assign a driver and display the booking details.
 8. If not available, ask the customer to choose another cab.
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5. Uniform Cost Search (UCS)

Algorithm

Uniform Cost Search (UCS) expands the node with the **lowest cumulative path cost** first. It guarantees the **least-cost path** from the start node to the goal node.

UCS Search Steps

Step Expanded Node Path Cost

1	S	0
2	A	1
3	C	2
4	D	3
5	G	4

Cost Calculation

- $S \rightarrow A = 1$
- $A \rightarrow C = 1$
- $C \rightarrow G = 2$

Total Cost = $1 + 1 + 2 = 4$

Least-Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

4

Result

Using the **Uniform Cost Search (UCS)** algorithm, the **minimum-cost path** from S to G is:

$S \rightarrow A \rightarrow C \rightarrow G$

Minimum Cost = 4